Keshav B. Deoskar

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Education

October 2026 – Ongoing
University of Sydney
Ph.D. Candidate, School of Physics
- Advisors: Dr. Dominic Williamson and Prof. Stephen Bartlett

August 2022 - May 2026
University of California, Berkeley
B.S. Engineering Physics

Research and Work Experience

2025-2026
Undergraduate Research
Prof. Joel Moore's group
- Condensed matter theory: Hopf insulators and novel band topologies.

2025-2026
Undergraduate Research
Prof. James Analytis' group
- Condensed matter experiment: Materials Synthesis, Characterization, and Data Analysis for flat-band physics in Kagome Lattice materials.
- Lab affiliate at **Lawrence Berkeley National Lab** during Summer 2025.

Summer 2023
Software Development Intern
Indus Novateur, SAP Gold Business
- Backend scripts for interfacing with SAP Databases.

Honors & Awards

- USYD Physics Foundation Scholarship, 2026
- Berkeley Physics Undergraduate Research Scholarship; Fall 2025, Spring 2026
- Discovery Undergraduate Research Scholar, Kavli Energy NanoScience Institute (ENSI); Spring 2025

Publications

2026
1. Yamakawa, K., Pentz, H. K., Rouquette, L., **Deoskar, K. B.**, Ray, A. & Analytis, J. Synthesis-driven Kagome lattice materials discovery towards many-body physics. *In Preparation* (2026).

Posters

April 2026
Automated Screening of Kagome Lattice Materials Towards Flat-Band Physics *Kohtaro Yamakawa, James Analytis, Lex Rouquette, Keshav Deoskar 2026 MRS Spring Meeting & Exhibit*

Teaching Experience

- Physics 105 - Analytical Mechanics, Reader; Jan 2026 - May 2026
- Physics/EECS/Chemistry C191A - Introduction to Quantum Computation, Reader; August 2025 - Dec 2025
- UC Berkeley Student Learning Center, Lower Division Mathematics tutor; August 2023 - May 2024

Outreach & Service

- QIP 2027 Reviewer
- Society of Physics Students (SPS), UC Berkeley Chapter - Officer (2024-2026)